

Assignment No. 9

M T W T F S S
Page No.:
Date: 14-6-26
YOUVA

Selection Sort

0 ps: 00 ()

```
int arr = {18, 22, 20, 19, 21};
```

```
int n = arr.length;
```

```
for (int i = 0; i < n - 1; i++)
```

```
    int minIdx = i;
```

```
    for (int j = i + 1; j < n; j++)
```

```
        if (arr[j] < arr[minIdx])
```

```
            minIdx = j;
```

```
    }
```

```
}
```

```
int temp = arr[minIdx];
```

```
arr[minIdx] = arr[i];
```

```
arr[i] = temp;
```

```
}
```

```
for (int i = 0; i < n; i++)
```

```
    S.O.P(arr[i] + " ");
```

```
}
```

```
}
```

run

```
arr = {18, 22, 20, 19, 21};
```


0 < 4

minIdx = 0

j = 1, 2
i = 0

1 < 5

22 < 18 x

2 < 5

20 < 18 x

temp = 18

3 < 5

19 < 18 x

4 < 5

21 < 18 x

temp = 18

arr[minIdx] = 18

arr[i] = 18

[18, 22, 20, 19, 21]

1 < 4

minIdx = 1

minIdx = x, 2, 3

2 < 5

20 < 22 ✓

minIdx = j

19 < 20 ✓

minIdx = j

21 < 19 x

temp = 19

arr[minIdx] = 22

arr[i] = temp

[18, 19, 20, 22, 21]

2 < 4

minIdx = 2

1 < 5

22 < 20 x

21 < 20 x

temp = 20

arr[minIdx] = 20

arr[j] = 20

[18, 19, 20, 22, 21]

3 < 4

minIdx = 3

4 < 5

21 < 22 ✓

minIdx = 4

temp = 21

arr[minIdx] = 22

arr[i] = 21

[18, 19, 20, 21, 22]

[35, 12, 56, 45, 23]

0 < 4

minIdx = 0

1 < 5

12 < 35 ✓

minIdx = 1

minIdx = 0, 1

2 < 5
→
56 < 12x

→ 3 < 5
45 < 12x

→ 4 < 5
23 < 12x

temp = 12
arr[minIdx] = 34
arr[i] = 13

[12, 34, 56, 45, 23]
0 1 2 3 4

1 < 4

→
minIdx = 1

→ 2 < 5 minIdx = 1, 4

→
56 < 34x

→ 3 < 5
→ 45 < 34x

→ 4 < 5
→ 23 < 34✓
minIdx = 4

temp = 23
arr[minIdx] = 34
arr[i] = 23

[12, 23, 56, 45, 34]
0 1 2 3 4

2 < 4

→
minIdx = 2

→ 3 < 5

→

45 < 56✓
minIdx = 3

→ 4 < 5

34 < 56

minIdx = 4

minIdx = 2, 3, 4

temp = 34

arr[minIdx] = 56

arr[i] = 34

[12, 23, 34, 45, 56]
0 1 2 3 4

3 < 4

→
minIdx = 3

→ 4 < 5

56 < 45x

0/p

[12, 23, 34, 45, 56]

Q2

Bubble Sort

0 psun 11}

int[] arr = {30, 25, 27, 35, 29}

int n = arr.length;

for(int i=0; i<n-1; i++){

boolean swapped = false;

for(int j=0; j<n-i-1; j++){

if(arr[j] > arr[j+1]){

int temp = arr[j];

arr[j] = arr[j+1];

arr[j+1] = temp;

swapped = true;

}

}

if(!swapped){

break;

}

}

}

Adityan

{30, 25, 27, 35, 29}

0 < 4

swapped = false, true

0 < 4

30 > 25 ✓

temp = 30

arr[j] = 25

arr[j+1] = temp

swapped = true

[25, 30, 27, 35, 29]

1 < 4

swapped = true

30 > 27 ✓

temp = 30

arr[j] = arr[j+1] = 27

arr[j+1] = 30

swapped = true.

[25, 27, 30, 35, 29]

2 < 4

30 > 35 x

30 > 35 x

3 < 4

35 > 29 ✓

temp = 35

arr[j] = 29

arr[j+1] = 35

[25, 27, 30, 29, 35]

1 < 4

1 < 3

27 > 30 x

2 < 3
→ 30 > 29

temp = 30
arr[j] = 29
arr[j+1] = 30
Swapped = true

O/p: [25, 27, 29, 30, 35]
0 1 2 3 4

3 < 4
→
Swapped = false

0 < 2
→ 25 > 27

1 < 2
→ 27 > 29 × O/p [25, 27, 29, 30, 35]

② [15, 10, 25, 40, 30]
0 1 2 3 4

0 < 4
→
Swapped = true
0 < 4
→ 15 > 10

temp = 15
arr[j] = 10
arr[j+1] = 15
Swapped = true

[10, 15, 25, 40, 30]
0 1 2 3 4

1 < 4
→ 15 > 25 ×

2 < 4
→ 25 > 40 ×

3 < 4
→ 40 > 30

temp = 40
arr[j] = 30
arr[j+1] = 40

[10, 15, 25, 30, 40]
0 1 2 3 4

1 < 4
→
Swapped = false
0 < 3
→ 10 > 15 ×

2 < 3
→ 15 > 25 ×

2 < 3
→ 25 > 30 ×

Swapped = false // break

O/p: [10, 15, 25, 30, 40]

Q3 Insert Sort

① psum (13)

int[] arr = {18, 22, 20, 19, 21}

for (int i = 1; i < arr.length; i++)
 int currentElement = arr[i];
 int left = i - 1;

while (left >= 0 && arr[left] > currentElement)

arr[left + 1] = arr[left];
 left--;

arr[left + 1] = currentElement;

3

3

Array {18, 22, 20, 19, 21}

1 < 5

0 >= 0 && 18 > 22 x

currentElement = 22, 20, 19, 21
 left = 0, 22, 19, 20, 21, 22

2 < 5

22 >= 0 && 22 > 20

arr[left + 1] = arr[left];
 left--;

[18, 22, 22, 19, 21]

0 >= 0 && 18 > 20 x

arr[left + 1] = 20

[18, 20, 22, 19, 21]

3 < 5

2 >= 0 && 22 > 19

arr[left + 1] = arr[left];
 left--;

[18, 20, 22, 22, 21]

1 >= 0 && 20 > 19 ✓

arr[left + 1] = arr[left];
 left--;

[18, 20, 20, 22, 21]

0 >= 0 && 18 > 19 x

arr[left + 1] = 19

[18, 19, 20, 22, 21]

4 < 5

3 >= 0 && 22 > 21 ✓

$$3+1=4$$

$$arr[1, 1+1] = 22$$

$$left = -$$

[18, 19, 20, 22, 22]
 0 1 2 3 4

$$2 > 0 + 1 \quad 20 > 21 \times$$

$$arr[3] = 21$$

Op [18, 19, 20, 21, 22]

Q [34, 12, 56, 45, 23]
 0 1 2 3 4

$$1 < 5$$



$$0 > 0 + 1 \quad 34 > 12$$

$$arr[left+1] = 34$$

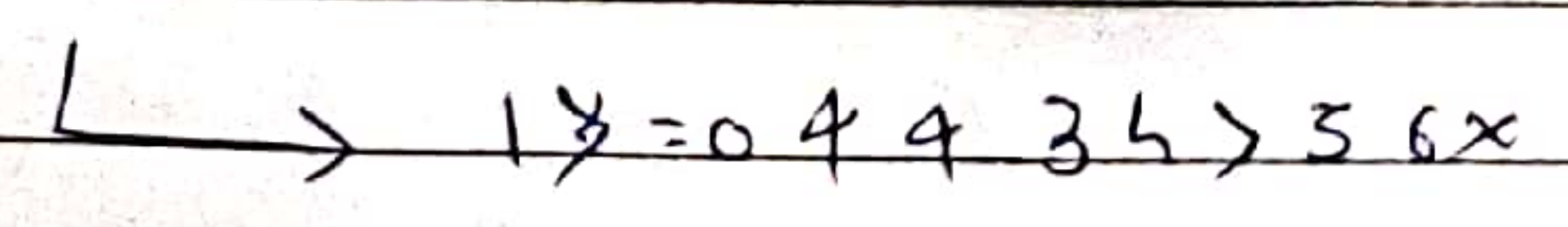
left = -

[34, 34, 56, 45, 23]
 0 1 2 3 4

$$arr[1] = 12$$

[12, 34, 56, 45, 23]
 0 1 2 3 4

$$2 < 5$$



$$3 < 5$$



$$2 > 0 + 1 \quad 56 > 45 \checkmark$$

$$arr[3] = 56$$

left = -

[12, 34, 56, 56, 23]
 0 1 2 3 4

$$1 > 0 + 1 \quad 34 > 45 \times$$

$$arr[left+1] = 45$$

[12, 34, 45, 56, 23]
 0 1 2 3 4

$$4 < 5$$



$$3 > 0 + 1 \quad 56 > 23$$

$$arr[left+1] = 56$$

left = -

[12, 34, 45, 56, 56]
 0 1 2 3 4

$$2 > 0 + 1 \quad 45 > 23$$

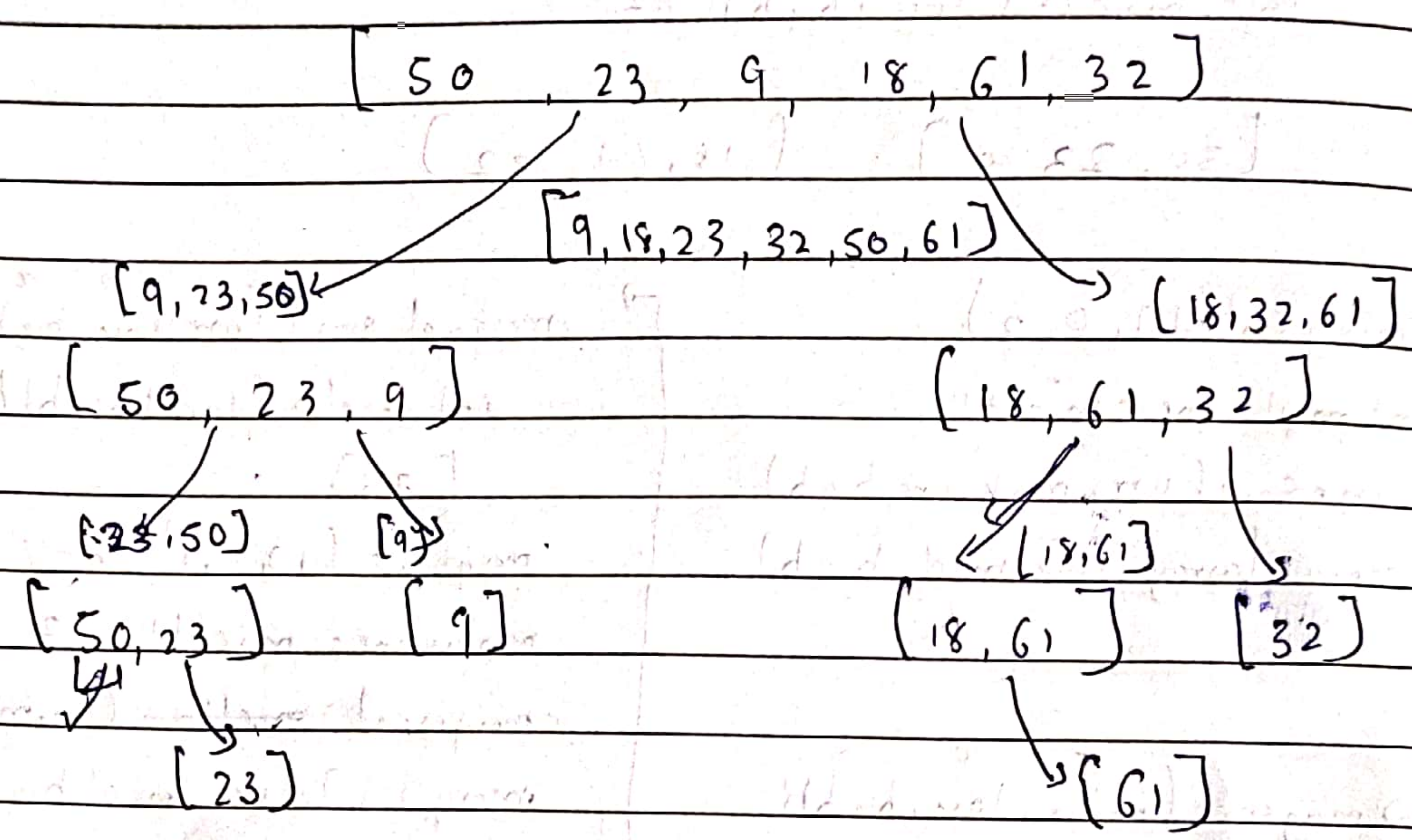
$$arr[left+1] = 45$$

left = -

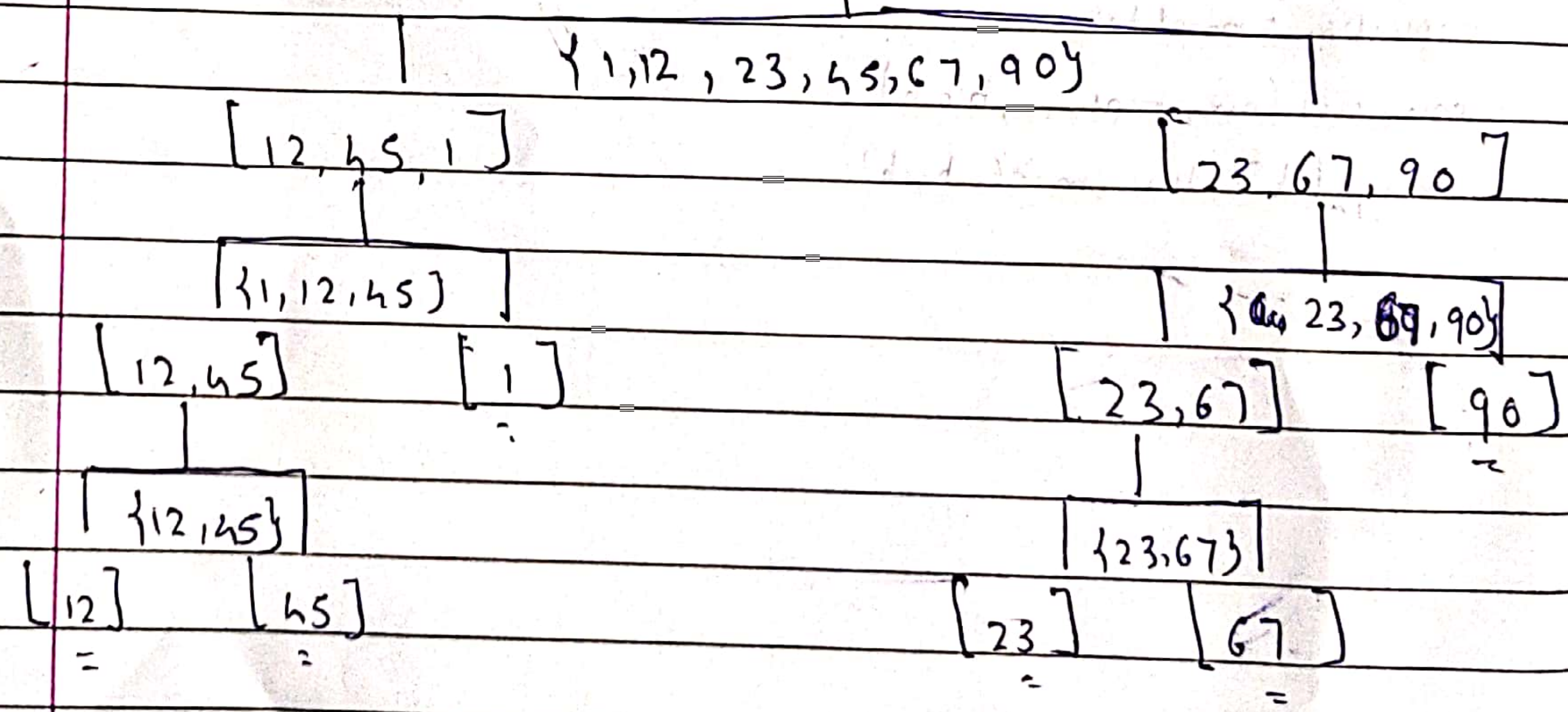
[12, 34, 45, 45, 56]
 0 1 2 3 4

Q. Merged Sort

① [50, 23, 9, 18, 61, 32]

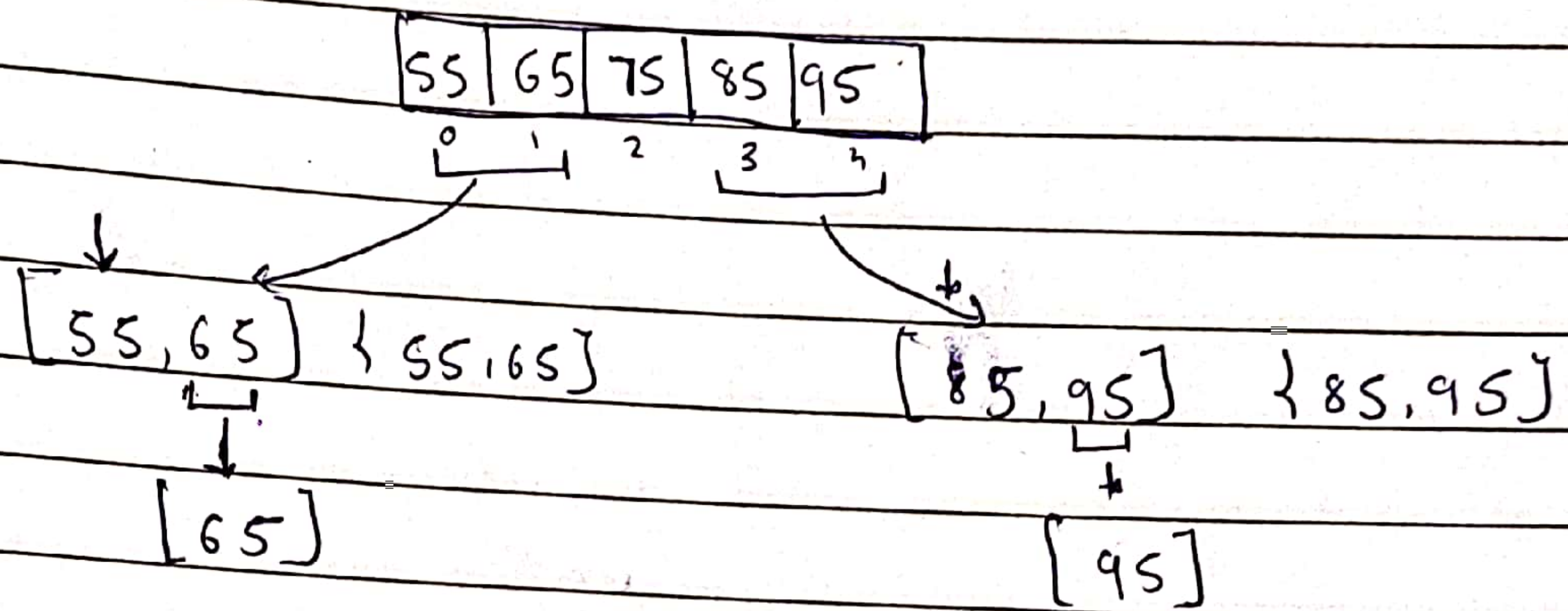


② [12, 45, 1, 23, 67, 90]



Q.6. # Quick Sort.

① $[75, 85, 55, 95, 65]$ $\{55, 65, 75, 85, 95\}$



② $[42, 56, 12, 67, 34]$ $\{12, 34, 42, 56, 67\}$

